Network Layer: Logical Addressing

The Network Layer is the third layer of the OSI model responsible for logical addressing, routing, and end-to-end packet delivery across interconnected networks.

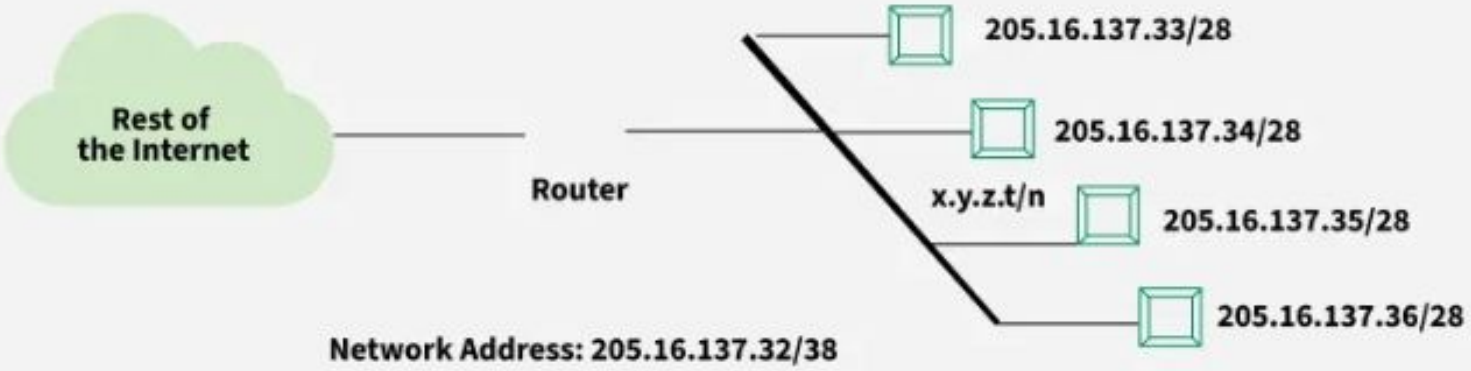
- Handles logical (IP) addressing of devices
- Determines optimal routing paths between networks
- Enables host-to-host communication across multiple networks

Functions of Network Layer

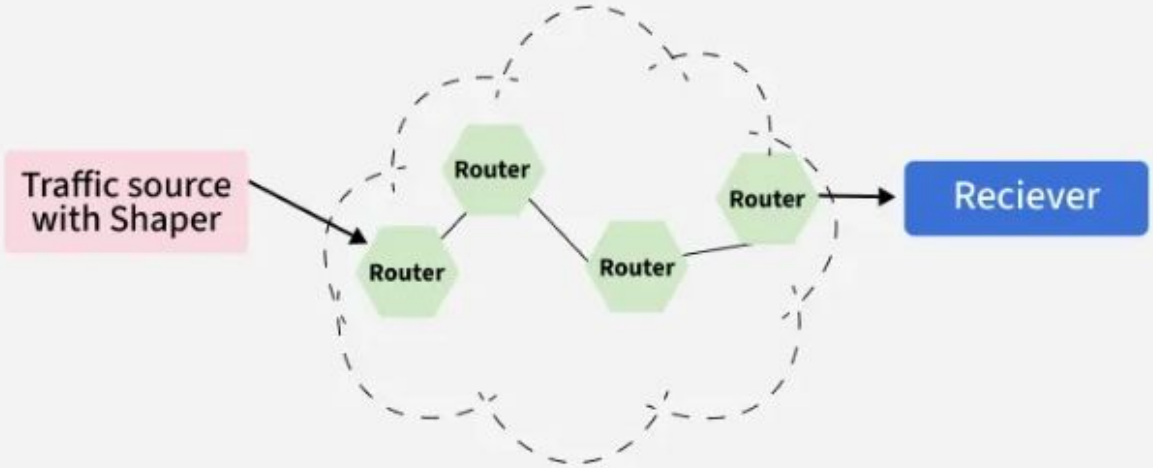
1. Assigning Logical Address
2. Packetizing
3. Host-to-host delivery
4. Forwarding
5. Fragmentation and Reassembly of packets
6. Logical Subnetting
7. Network Address Translation
8. Routing

1. Assigning Logical Address

It assigns unique IP addresses to devices for identification across networks.

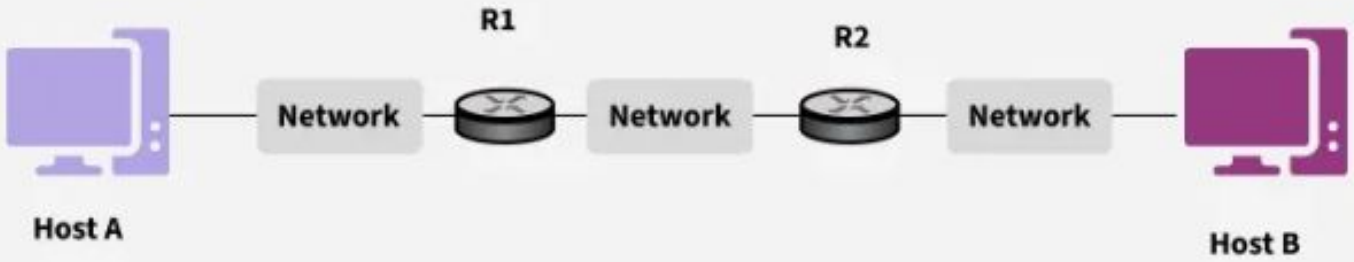


2. Packetizing



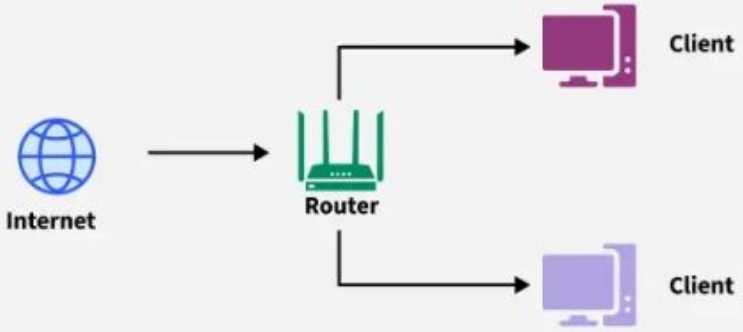
3. Host-to-host delivery

It ensures that data is reliably transmitted between two hosts across a network

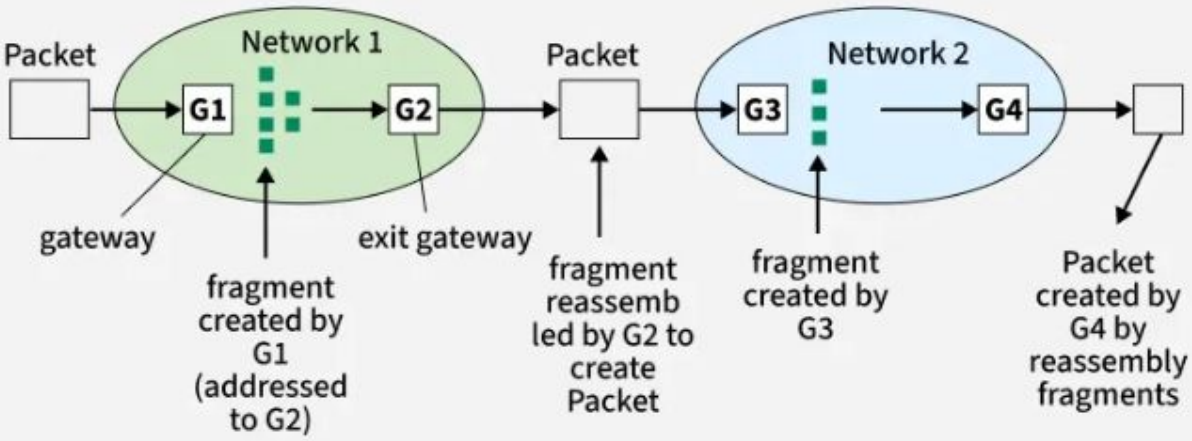


4.Forwarding

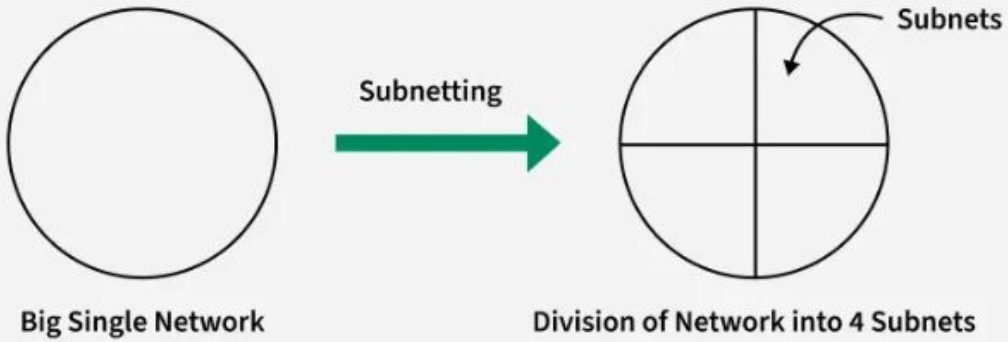
Forwarding moves packets through a router to the correct outgoing interface based on their destination.



5. Fragmentation and Reassembly of packets

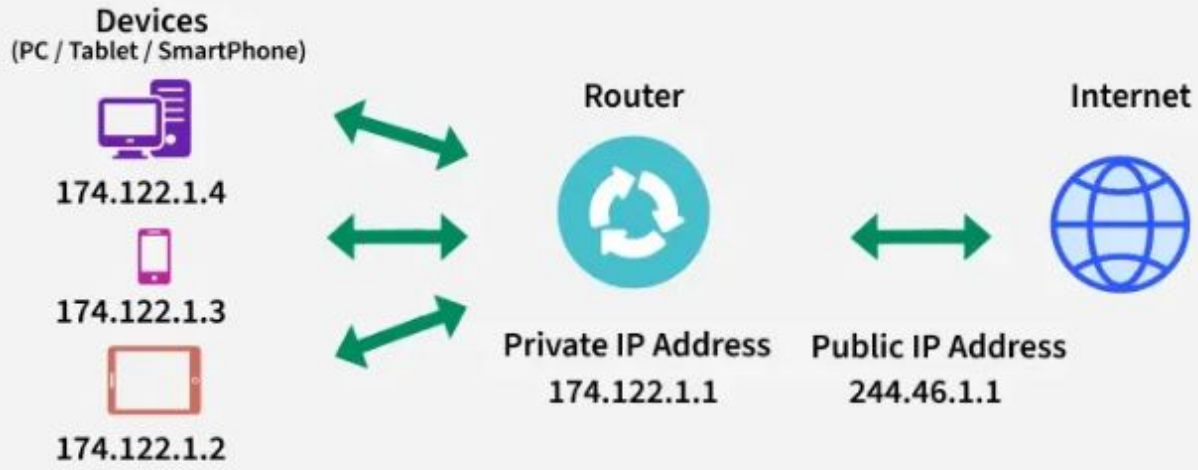


6. Logical Subnetting

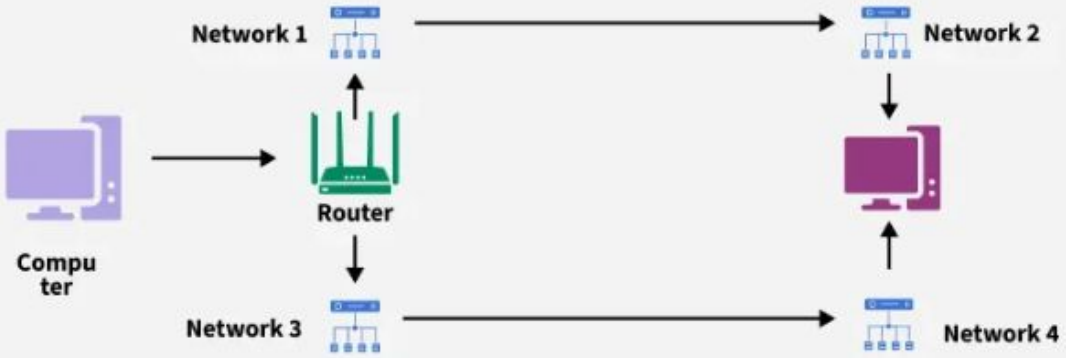


7. Network Address Translation

NAT is a method of mapping private IP addresses within a local network to a single public IP address (and vice versa)



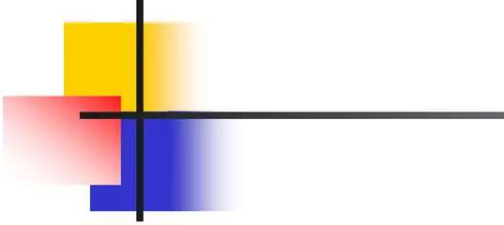
8. Routing



19-1 IPv4 ADDRESSES

An IPv4 address is a 32-bit address that uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet.

Address Space
Notations
Classful Addressing
Classless Addressing
Network Address Translation (NAT)



Note

An IPv4 address is 32 bits long.

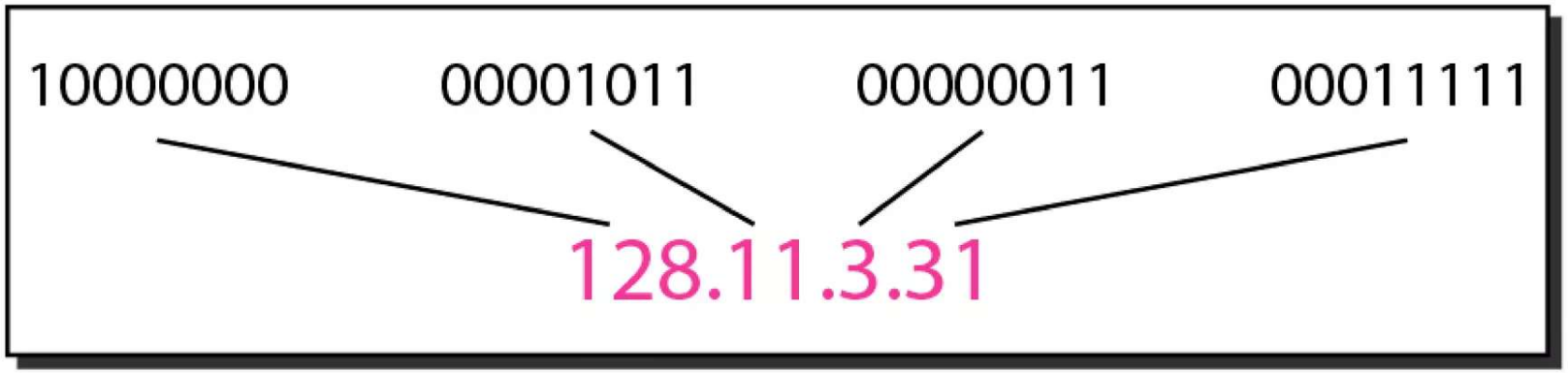
Note

The IPv4 addresses are unique and universal.

Note

The address space of IPv4 is 2^{32} or 4,294,967,296.

Figure 19.1 *Dotted-decimal notation and binary notation for an IPv4 address*



Example 19.1

Change the following IP addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11111001 10011011 11111011 00001111

Solution

We replace each group of 8 bits with its equivalent decimal number and add dots for separation:

a. 129.11.11.239

b. 249.155.251.15

Example 19.2

Change the following IP addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 75.45.34.78

Solution

We replace each decimal number with its binary equivalent :

a. 01101111 00111000 00101101 01001110

b. 01001011 00101101 00100010 01001110

Types of Addressing

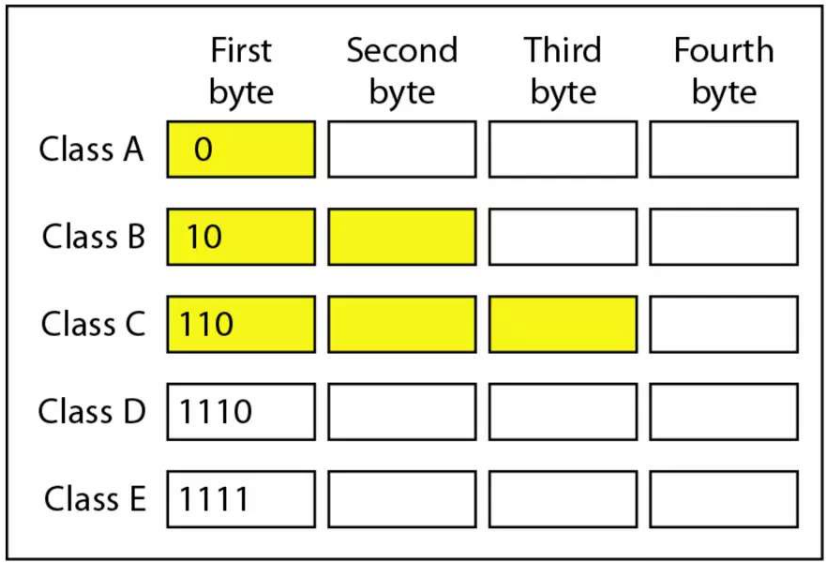
There are two type of addressing. They are

- Classful addressing
- Classless addressing

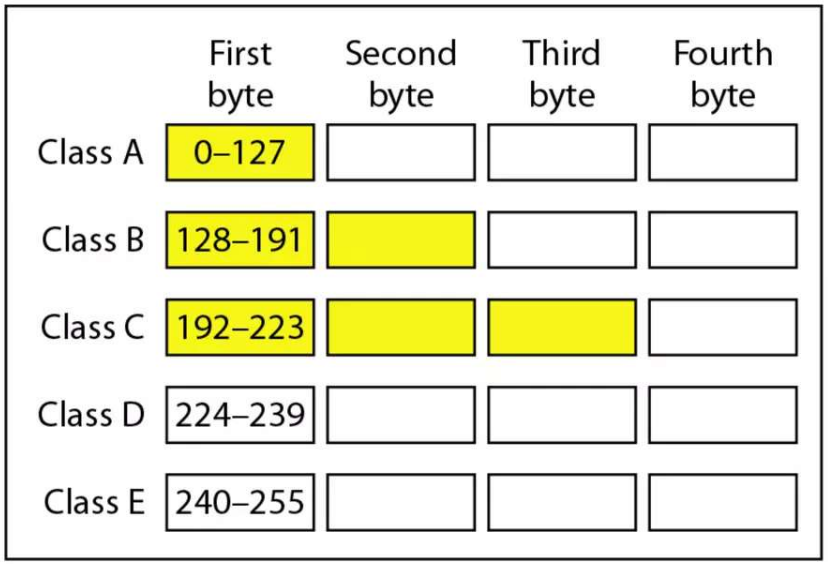
Note

In classful addressing, the address space is divided into five classes: A, B, C, D, and E.

Figure 19.2 *Finding the classes in binary and dotted-decimal notation*



a. Binary notation



b. Dotted-decimal notation

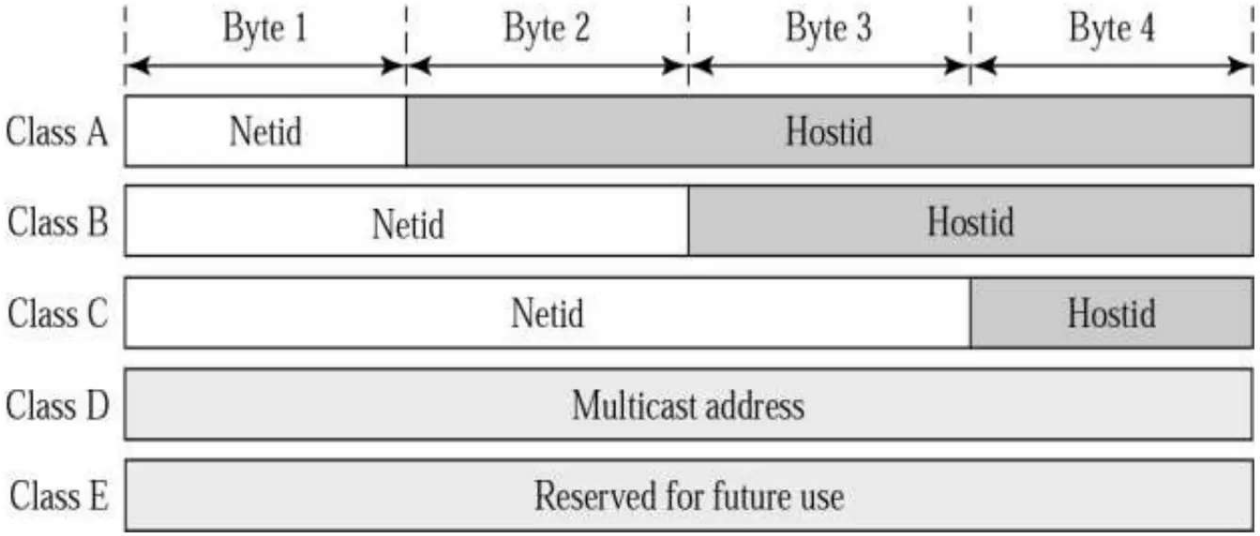
- Class A address: designed for large organizations with a large number of attached hosts or routers. (Wasted and not used)
- Class B address: designed for midsize organizations with tens of thousands of attached hosts or routers (too large for many organizations)
- Class C address: designed for small organizations with a small number of attached hosts or routers. (too small for many organizations)
- Class D address: designed for multicasting. (Waste of addresses)
- Class E address: reserved for future use (waste of addresses)

The following IP address ranges belong to **Google**:

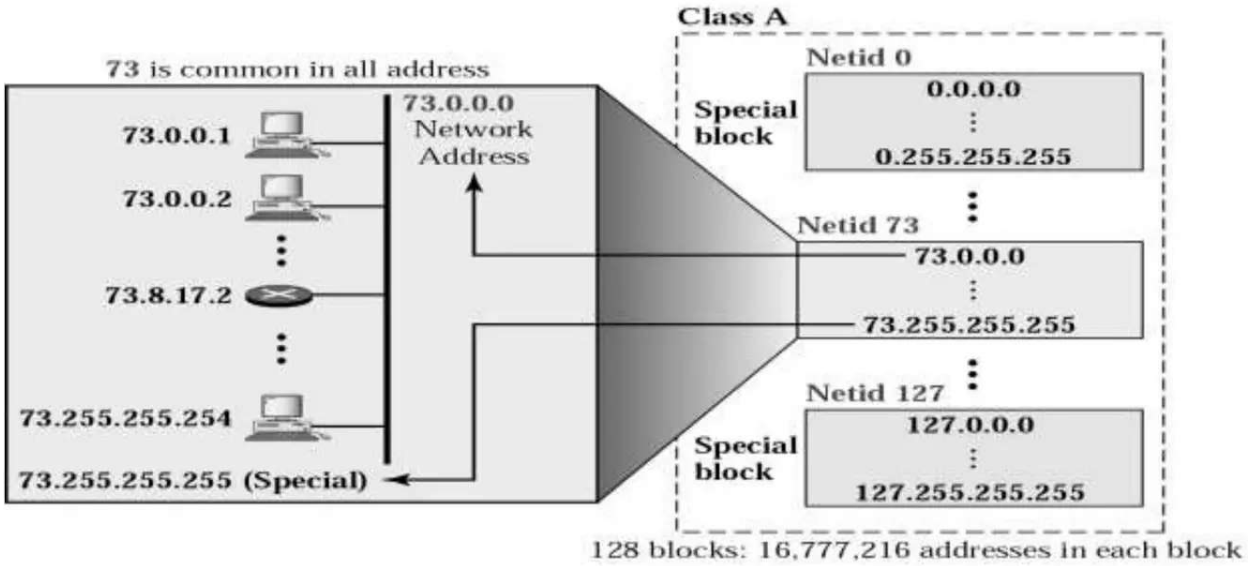
- 64.233.160.0 - 64.233.191.255
- 66.102.0.0 - 66.102.15.255
- 66.249.64.0 - 66.249.95.255
- 72.14.192.0 - 72.14.255.255
- 74.125.0.0 - 74.125.255.255
- 209.85.128.0 - 209.85.255.255
- 216.239.32.0 - 216.239.63.255

Netid and Hostid

- The address is divided into netid and hostid.
- These parts are of varying lengths, depending on the class.



Blocks in class A



Note

In classful addressing, a large part of the available addresses were wasted.

Example 19.4

Find the class of each address.

a. 00000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

c. 14.23.120.8

d. 252.5.15.111

Solution

a. The first bit is 0. This is a class A address.

b. The first 2 bits are 1; the third bit is 0. This is a class C address.

c. The first byte is 14; the class is A.

d. The first byte is 252; the class is E.

Table 19.1 *Number of blocks and block size in classful IPv4 addressing*

Class	*Number of Blocks*	*Block Size*	*Application*
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

Mask

- It can help us to find the netid and hostid.
- It is 32-bit number made of contiguous 1s followed by contiguous 0s.

Table 19.2 *Default masks for classful addressing*

Class	*Binary*	*Dotted-Decimal*	*CIDR*
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255.0	/24

CIDR (Classless Interdomain Routing) or slash notation:

- It is used to show the mask in the form /n (n=8, 16, 24)
- In classful addressing, a large part of the available addresses were wasted.

Sub netting

- If an organization was granted a large block in class A or B, it could divide the addresses into several contiguous groups and assign each group to smaller networks (subnets) .
- It increases the number of 1s in the mask.

Super netting

- Although class A and B addresses are almost depleted, class C addresses are still available (size of block= 256 so address did not satisfy the needs).
- In super netting, an organization can combine several class C blocks to create a larger range of addresses.
- Several networks are combined to create a super network (**super net**). E.g. Organization needs 1000 address can be granted 4 contiguous class C blocks. Create one super network. It decreases the number of 1s in the mask.

Address Depletion

- Class C block is too small for most mid size organizations. So the solution is Classless addressing.
- Classful addressing, which is almost obsolete, is replaced with classless addressing.

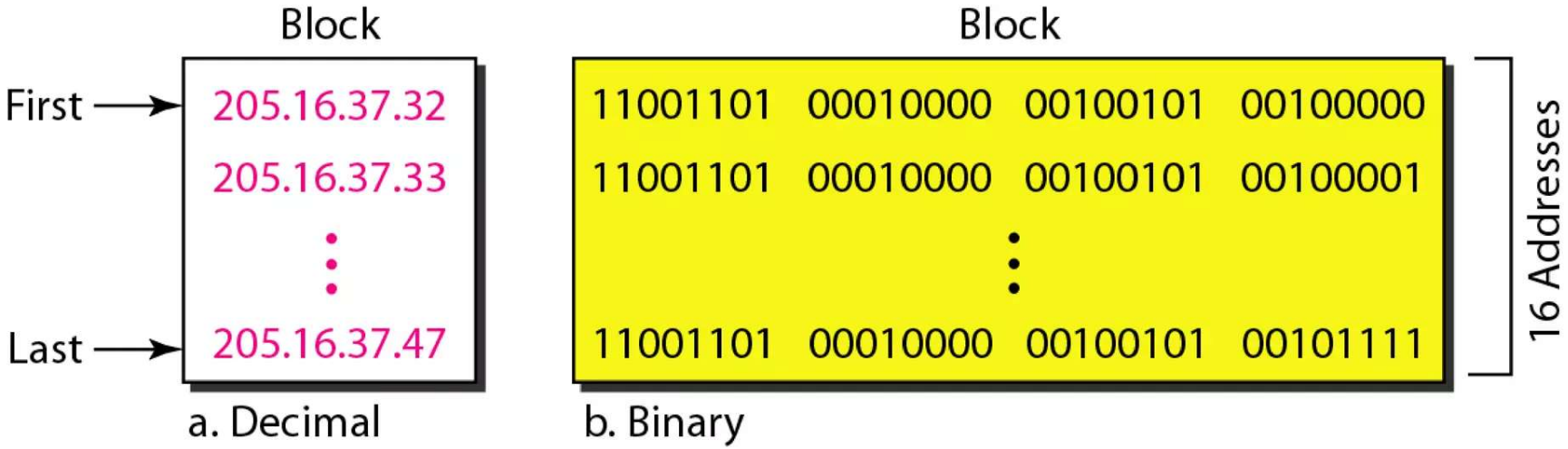
Note

Classful addressing, which is almost obsolete, is replaced with classless addressing.

Classless addressing

- To overcome address depletion and give more organizations access to the internet, classless addressing was designed.
- There are no classes, but the addresses are still granted in blocks.
- The size of the block (the number of addresses) varies based on the nature and size of the entity.
  - Household: 2 addresses
  - Large organization: thousands of addresses.
  - ISP: thousands or hundreds of thousands based on the number of customers it may serve.

Figure 19.3 *A block of 16 addresses granted to a small organization*



Note

In IPv4 addressing, a block of addresses can be defined as x.y.z.t /*n* in which x.y.z.t defines one of the addresses and the /*n* defines the mask.

Note

The first address in the block can be found by setting the rightmost $32 - n$ bits to 0s.

So Far we Study in IPv4

- Header
- Address format
  - Classful IP Address
    - Class A
      - 0 to 127
    - Class B
      - 128 to 191
    - Class C
      - 192 to 223
    - Class D
      - 224 to 239
    - Class E
      - 240 to 255

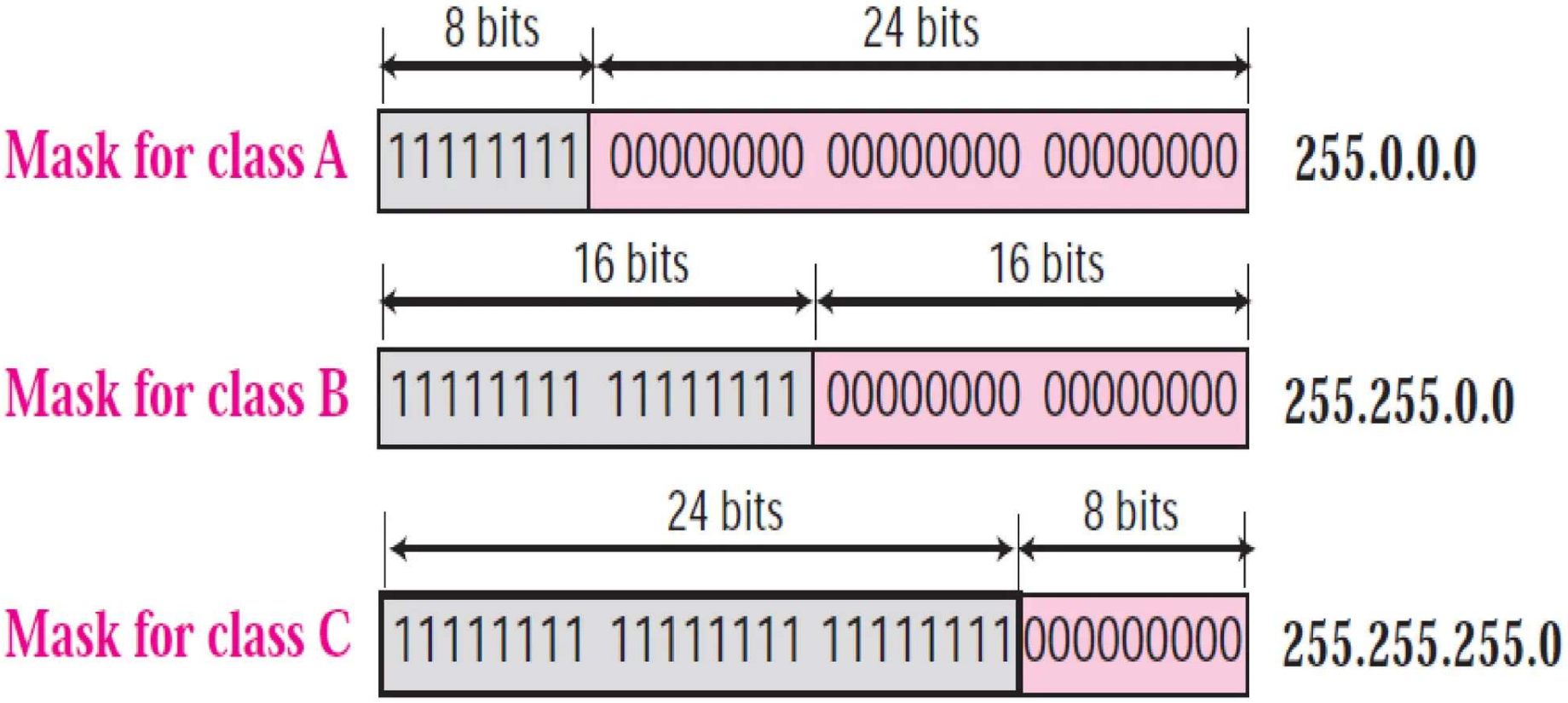
Routing

- The data communication is done through the Network id (1st part) of an IP Address.
- The data is reach first to the destination network (using Network id) and then to that particular host (Destination) in that network (using Host id).

Network Address and Network Mask

- **Network Address:** The First address of a Network is the network address of that network, which specify the network.
  - All the data communication in WAN is done through the Network address, so that the data first reach to network and then to Host.
- **Network Mask:** A network mask or a default mask in classful addressing with n leftmost bits all set to 1s and (32 − n) rightmost bits all set to 0s.

Default Network Masks



Network Address

- To extract the network address from the destination address of a packet, a router uses the AND operation.
- When the destination address (or any address in the block) is ANDed with the default mask, the result is the network address.
- **AND Operation:**
- 1 AND 1 = 1
- 1 AND 0 = 0
- 0 AND 1 = 0
- 0 AND 0 = 0

Example

- **Example:** A router receives a packet with the destination address 131.24.67.32. Show how the router finds the network address of the packet.
- **Solution:** Since the class of the address is B, the router applies the default mask for class B, 255.255.0.0 to find the network address.
- Dest. Add. : 10000011. 00011000. 01000011. 00100000

  AND
- Mask: 11111111.11111111.00000000.00000000
- Net. Add. : 10000011. 00011000. 00000000.00000000
- = 131.24.0.0

Example

- **Example:** A router receives a packet with the destination address 131.24.67.32. Show how the router finds the network address of the packet.
- **Solution:** Since the class of the address is B, the router applies the default mask for class B, 255.255.0.0 to find the network address.

Destination address	→	131	.	24	.	67	.	32
Default mask	→	255	.	255	.	0	.	0
Network address	→	131	.	24	.	0	.	0

Three-Level Addressing: Subnetting

- The IP addresses were originally designed with two levels of addressing.
- To reach a host on the Internet, we must first reach the network and then the host.
- It soon became clear that we need more than two hierarchical levels, for two reasons.
- First, an organization that was granted a block in class A or B needed to divide its large network into several subnetworks for better security and management.
- Second, the blocks in class A and B are larger and most of the IP address then lost so class A or B could divide the block into smaller subblocks.

Subnetting

- In subnetting, a network is divided into several smaller sub-networks (subnets)
- Each sub network having its own sub network address.
- For a network here we will use /n,
- Where the “n” will show the length of the network id.

Subnet Mask

- The network mask is use when a network is not subnetted.
- When we divide a network to several sub networks, we need to create a sub network mask (or subnet mask) for each sub network.
- A sub network has subnet id and host id.

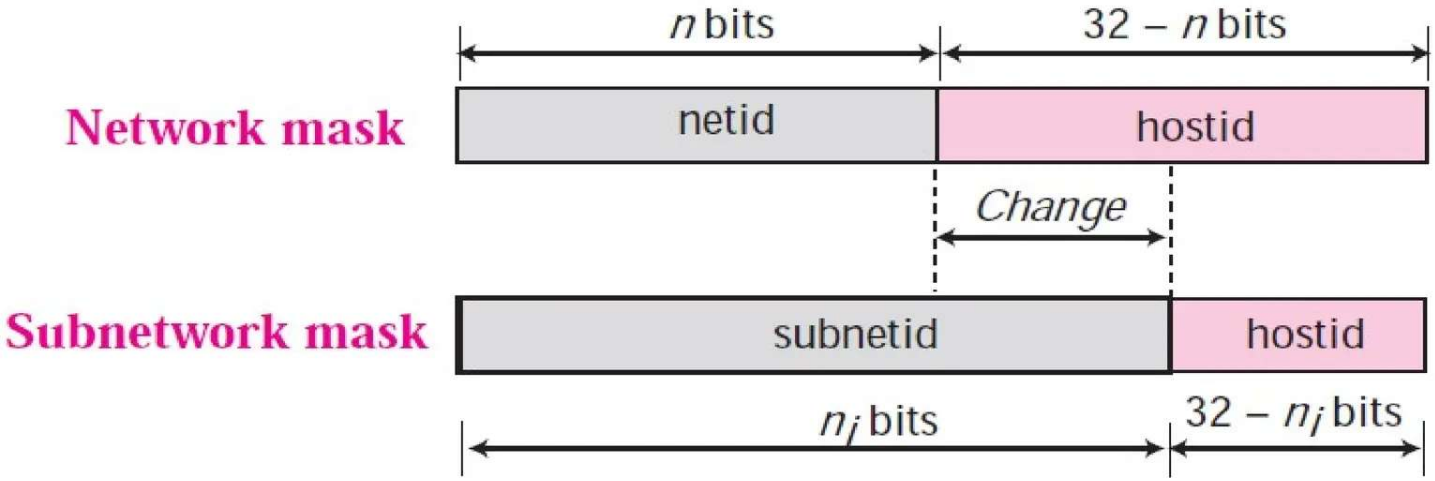
Subnet Mask Cont.

- Subnetting increases the length of the net id and decreases the length of host id.
- When we divide a network to *"s" number of subnetworks, each of equal numbers of* hosts,
- we can calculate the subnetid for each subnetwork.

Subnet Mask cont.

- For Example: Calculate a subnet mask for a network 141.14.0.0/16. we want to make 4 sub networks for this IP address uing subnet.
- First we have to calculate Subnet Mask.
- We need 4 networks so $2^2 = 4$. so we require 2 more bits to add them in Network Id bits, so now we have 16 + 2 = 18 network id bits.
- So Subnet mask for this class B will now =
- 11111111.11111111.11000000.00000000
- OR: 255.255.192.0

Subnet Mask cont.

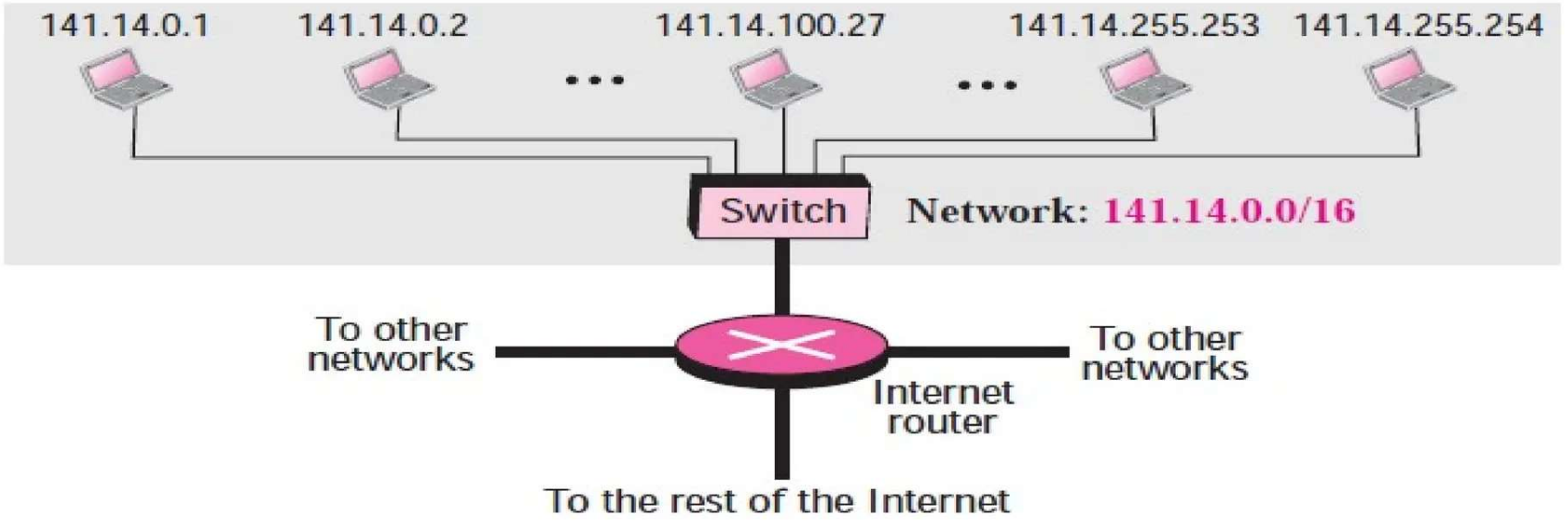


Subnetting Host IDs

- Now we can further subdivide a single IP to Sub networks through Subnetting.
- In Each Subnet the First and Last IP address will be reserved.
- First IP Address will be the **Network Address** for that small network (Sub Network).
- The Last IP address will be the **Broadcast Address** for that small network (Sub Network).
- It mean that the a sub network will always support to -2 sub host ids.

Example (with out subnetting)

- We have an IP address: 141.14.0.0
- It is Class B IP address.
- This network can have $2^{16} = 65536 - 2 = 65534$ Hosts.
- It is a single Network.



Example (Using Subnetting)

- We have an IP address: 141.14.0.0
- It is Class B IP address.
- This network can have $2^{16} = 65536 - 2 = 65534$ Hosts.
- But we want to have let suppose 4 sub networks of this single network
- As $2^n = 4$ if $n = 2$, so $2^2 = 4$
- So we require 2 extra bits to add in the network id bits.

Example Cont.

- IP address: 141.14.0.0
- Network ID: 141.14.0.0/16
- 4 Networks = 2 bits
- 141.14.0.0/18
- Now we have 18 bits for Network id and the remaining bits for host id = 32-18 = 14
- So now we can make 4 networks and each network will have $2^{14} = 16384 - 2 = 16382$ Hosts.

Example Cont.

- IP address: 141.14.0.0
- Binary : 10001101.00001110.00000000.0000000
- Network Mask for Class B: 255.255.0.0
- Or: 11111111.11111111.00000000.00000000
- **For Subnetting: for 4 sub networks**
- IP address: 141.14.0.0/18
- Subnet Mask: 11111111.11111111.11000000.00000000
- Or: 255.255.192.0

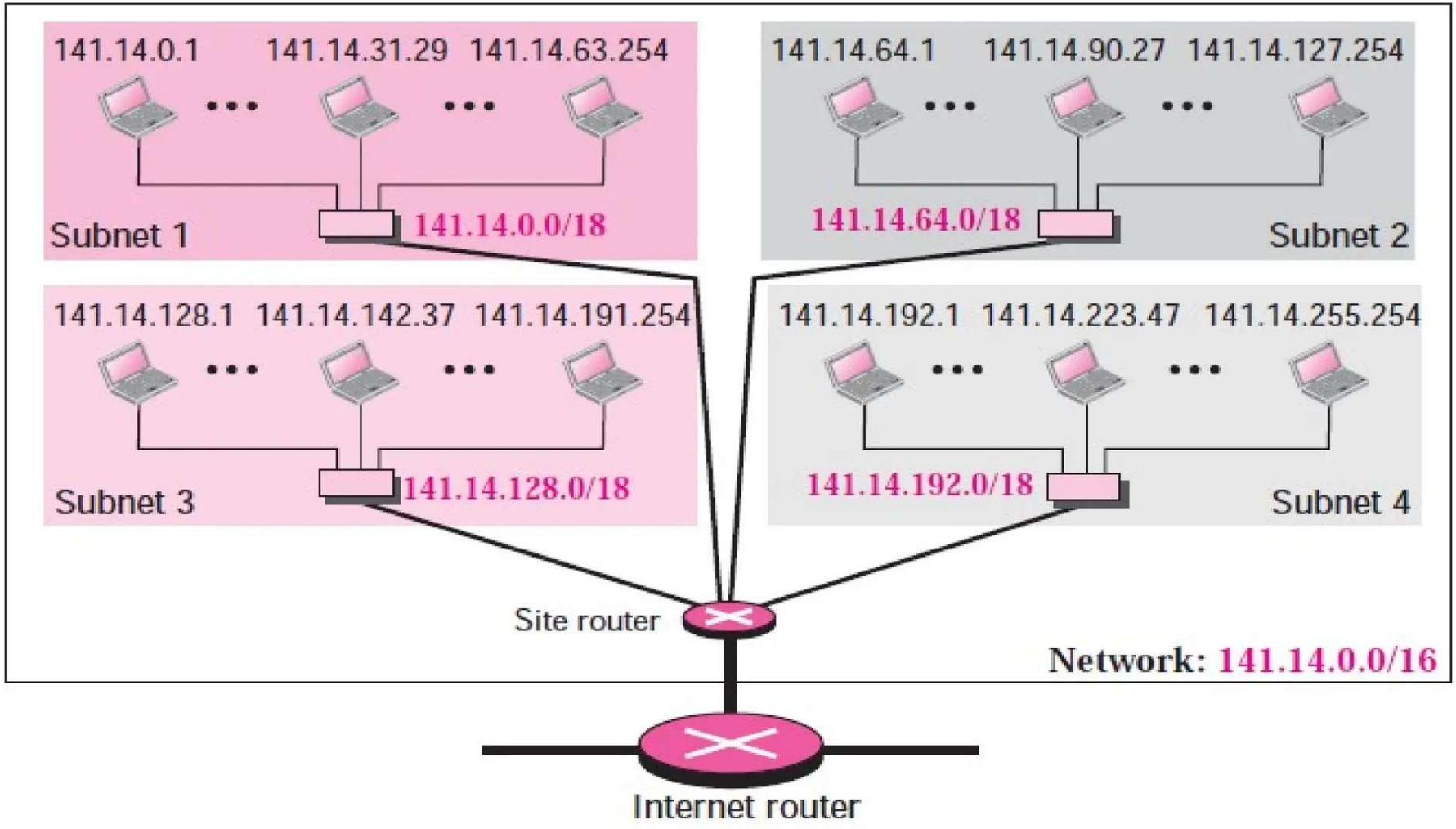
Example cont.

- **First Subnet:**
- IP address: 141.14.0.0/18
- Hosts in each network $2^{14} = 16384 - 2 = 16382$
- Range:
- 141.14.0.0 to 141.14.63.255
- 1st Subnet ID: 141.14.0.0/18

Example cont.

- **2nd Subnet:** Range:
- 141.14.64.0 to 141.14.127.255
- 2nd Subnet ID: 141.14.64.0/18
- **3rd Subnet:** Range:
- 141.14.128.0 to 141.14.191.255
- 3rd Subnet ID: 141.14.128.0/18
- **4th Subnet:** Range:
- 141.14.192.0 to 141.14.255.255
- 4th Subnet ID: 141.14.192.0/18

Example cont.



Example to Find the Subnet Address of a host IP

- A network is divided into four subnets. Since one of the addresses in a subnet is 141.14.120.77, Find the subnet address.
- **Sol:** As the IP is from Class B i.e. **141.14**.120.77
- Divided into 4 sub networks so 2 bits are reserved.
- So the subnet mask is
- 11111111.11111111.11000000.00000000
- Or: 255.255.192.0

Example Cont.

- IP address: 141.14.120.77
- Binary : 10001101.00001110.01111000.01001101
- Subnet mask: 255.255.192.0
- Or: 11111111.11111111.11000000.00000000
- Subnet Address: IP AND subnet mask:
- 10001101.00001110.01111000.01001101

  AND
- 11111111.11111111.11000000.00000000
- **Subnet Address:**
- **10001101.00001110.01000000.00000000**

  = 141.14.64.0

Example Cont.

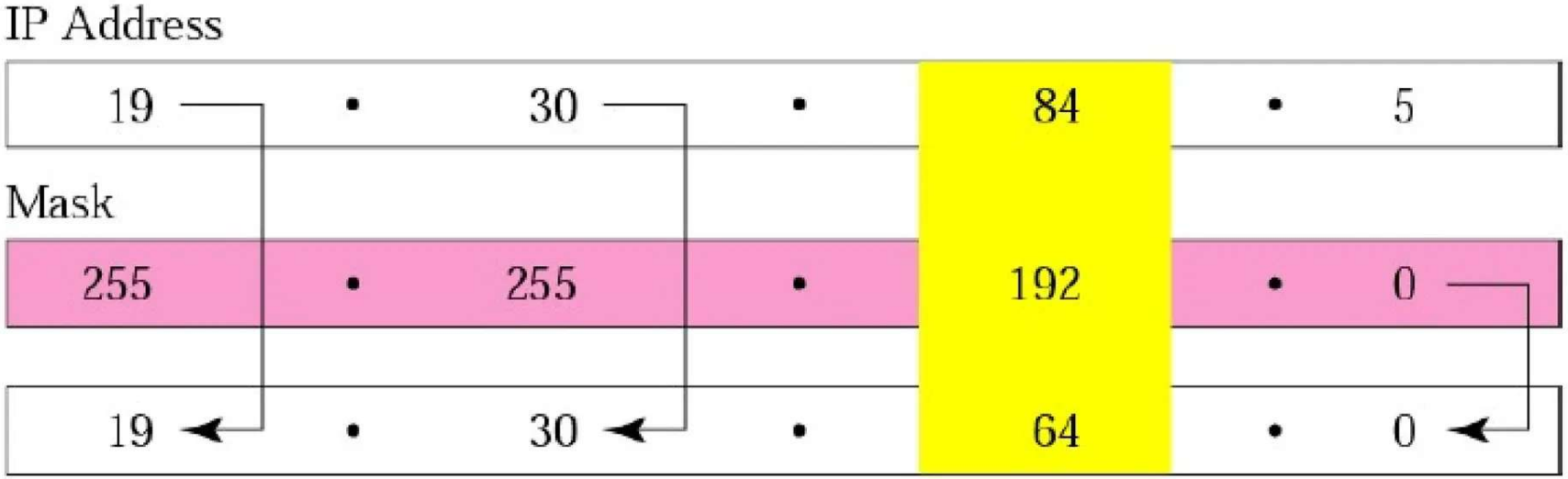
Address	→	141	.	14	.	120	.	77
Mask	→	255	.	255	.	192	.	0
Subnet Address	→	141	.	14	.	64	.	0

Address (120)	0	+	64	+	32	+	16	+	8	+	0	+	0	+	0
Mask (192)	128	+	64	+	0	+	0	+	0	+	0	+	0	+	0
Result (64)	0	+	64	+	0	+	0	+	0	+	0	+	0	+	0

Exercise

- What is the subnet address if the IP address is 19.30.84.5 and the mask is 255.255.192.0?

Solution



84	0	1	0	1	0	1	0	0
192	1	1	0	0	0	0	0	0
64	0	1	0	0	0	0	0	0

Exercise

- A company is granted the IP address 201.70.64.0, The company needs six subnets. Design the subnets.

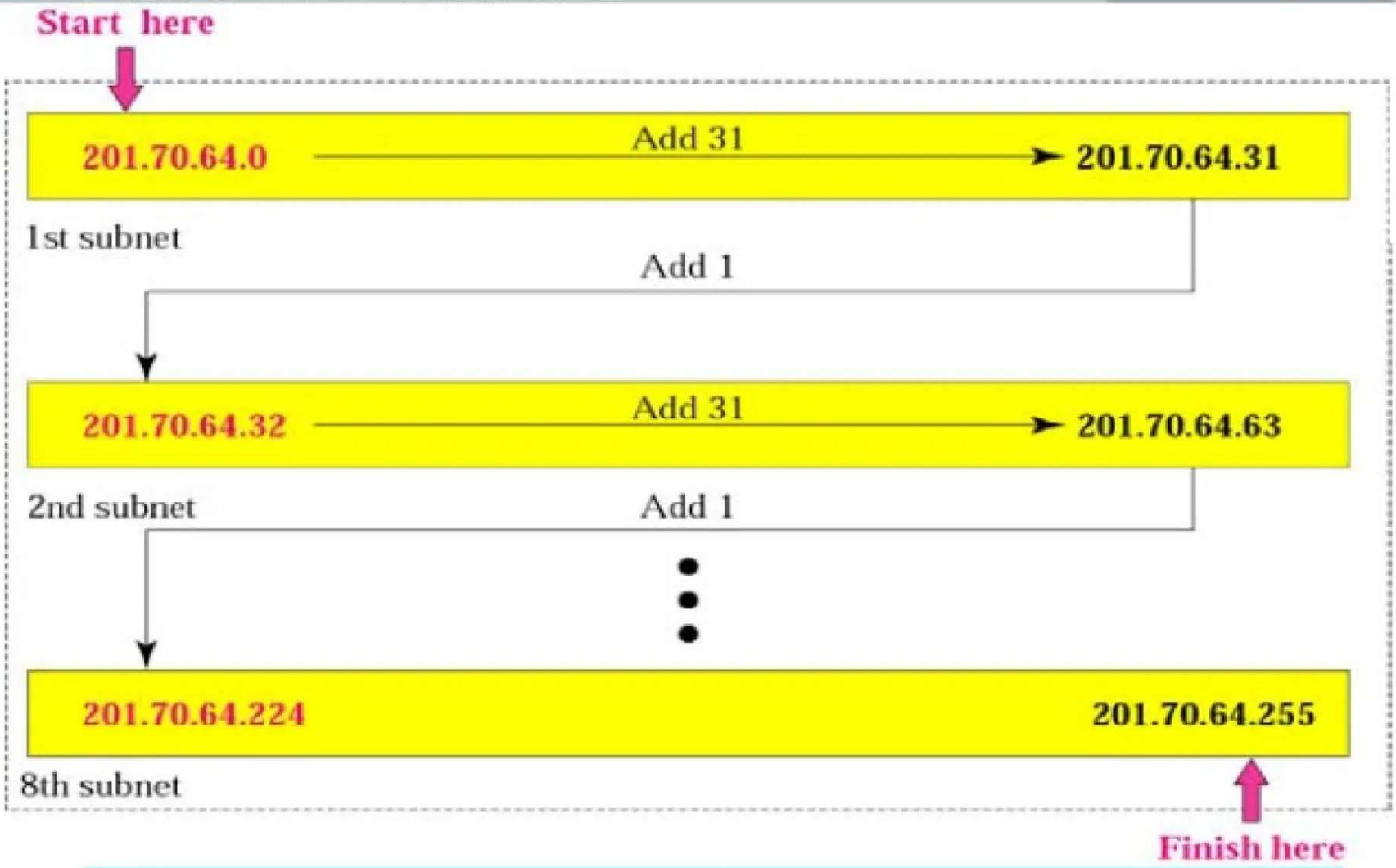
Solution

- The Address is of Class C: 201.70.64.0
- The company needs six subnets. This number 6 is not a power of 2. The next number that is a power of 2 is 8 (2^3).
- We need 3 more 1s in the subnet mask. The total number of 1s in the subnet mask is 27 (24 + 3).
- The total number of 0s is 5 (32 - 27). The mask is

  11111111 11111111 11111111 11100000

  Or 255.255.255.224
- The number of subnets is 8.
- The number of addresses in each subnet is 2^5 (5 is the number of 0s) = 32-2 = 30

Address Range



The subnet number is the first address in the subnet.

Exercise

- **A subnet has a subnet address of 141.14.64.0, find the address range in the subnets if its subnet masks are**

1. 255.255.224.0

2. 255.255.240.0

Solution

- **Subnet address : 141.14.64.0**
- **Class B:**

1. For 255.255.224.0

As 224 = 128 + 64 + 32

So 3 bits are Reserved for Network and 5 for host in 2nd Octat

Total Number of Hosts bits = 8 + 5 = 13

Total Number of Host = 2^{13} = 8192 – 2 = 8190

Solution cont.

- **Subnet address : 141.14.64.0**
- **Class B:**

2. For 255.255.240.0

As 240 = 128 + 64 + 32 + 16

So 4 bits are Reserved for Network and 4 for host in 2nd Octat

Total Number of Hosts bits = 8 + 4 = 12

Total Number of Host = 2^{12} = 4096 – 2 = 4092

Solution Cont.

If the network mask is 255.255.224.0 (netid - 16bits, subnetid - 3bits)
The address range is:

141.14	010	0 0000 0000 0000
netid	subnetid	hostid

to

141.14	010	1 1111 1111 1111

If the network mask is 255.255.240.0 (netid - 16bits, subnetid - 4bits)
The address range is:

141.14	0100	0000 0000 0000
netid	subnetid	hostid

to

141.14	0100	1111 1111 1111

Example 19.6

A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address in the block?

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32−28 rightmost bits to 0, we get

11001101 00010000 00100101 0010000

or

205.16.37.32.

This is actually the block shown in Figure 19.3.

Note

The last address in the block can be found by setting the rightmost 32 − n bits to 1s.

Example 19.7

Find the last address for the block in Example 19.6.

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32 − 28 rightmost bits to 1, we get

11001101 00010000 00100101 00101111

or

205.16.37.47

This is actually the block shown in Figure 19.3.

Note

The number of addresses in the block can be found by using the formula 2^{32-n}.

Example 19.8

Find the number of addresses in Example 19.6.

Solution

The value of n is 28, which means that number of addresses is 2^{32-28} *or 16.*

Example 19.9

Another way to find the first address, the last address, and the number of addresses is to represent the mask as a 32-bit binary (or 8-digit hexadecimal) number. This is particularly useful when we are writing a program to find these pieces of information. In Example 19.5 the /28 can be represented as

11111111 11111111 11111111 11110000

(twenty-eight 1s and four 0s).

Find

a. The first address

b. The last address

c. The number of addresses.

Example 19.9 (continued)

Solution

a. ***The first address can be found by ANDing the given addresses with the mask. ANDing here is done bit by bit. The result of ANDing 2 bits is 1 if both bits are 1s; the result is 0 otherwise.***

Address:	11001101	00010000	00100101	00100111
Mask:	**11111111**	**11111111**	**11111111**	**11110000**
First address:	11001101	00010000	00100101	00100000

Example 19.9 (continued)

b. ***The last address can be found by ORing the given addresses with the complement of the mask. ORing here is done bit by bit. The result of ORing 2 bits is 0 if both bits are 0s; the result is 1 otherwise. The complement of a number is found by changing each 1 to 0 and each 0 to 1.***

Address:	11001101	00010000	00100101	00100111
Mask complement:	**00000000**	**00000000**	**00000000**	**00001111**
Last address:	11001101	00010000	00100101	00101111

Note

NETWORK ADDRESS:
The first address in a block is normally not assigned to any device; it is used as the network address that represents the organization to the rest of the world.

Example:

- Given the address 23.56.7.91, find the network address.

Solution:

- The class is A. Only the first byte defines the netid. We can find the network address by replacing the hostid bytes (56.7.91) with 0s. Therefore, the network address is 23.0.0.0.

Example:

- Given the address
- 132.6.17.85 and 23.56.7.91, find the network address?

Solution

- 1. The class is B. The first 2 bytes defines the netid. We can find the network address by replacing the hostid bytes (17.85) with 0s. 132.6.0.0.
- 2. The class is A. Only the first byte defines the netid. We can find the network address by replacing the hostid bytes (56.7.91) with 0s. Therefore, the network address is 23.0.0.0.

19-2 IPv6 ADDRESSES

Despite all short-term solutions, address depletion is still a long-term problem for the Internet. This and other problems in the IP protocol itself have been the motivation for IPv6.

Topics discussed in this section:

Structure
Address Space

Note

An IPv6 address is 128 bits long.

Figure 19.14 *IPv6 address in binary and hexadecimal colon notation*

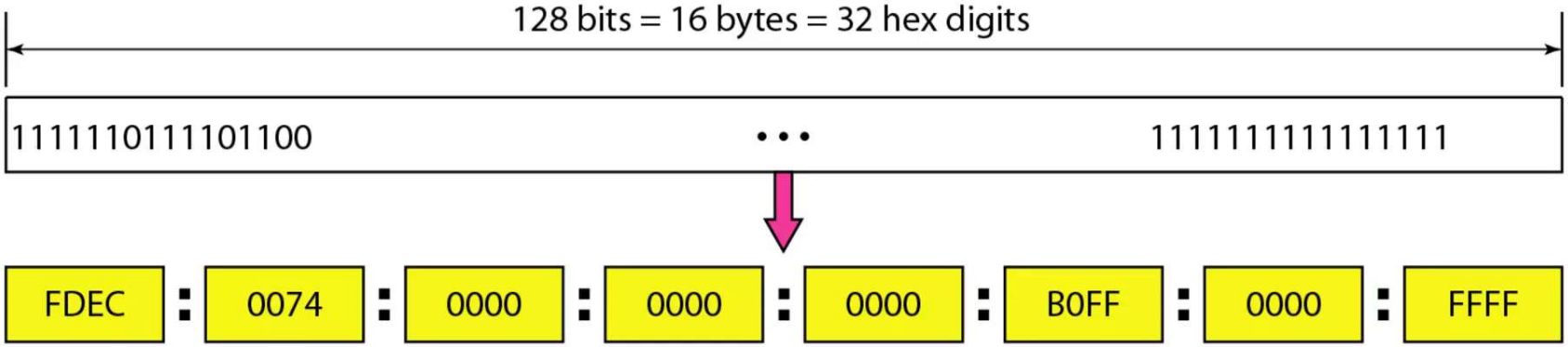
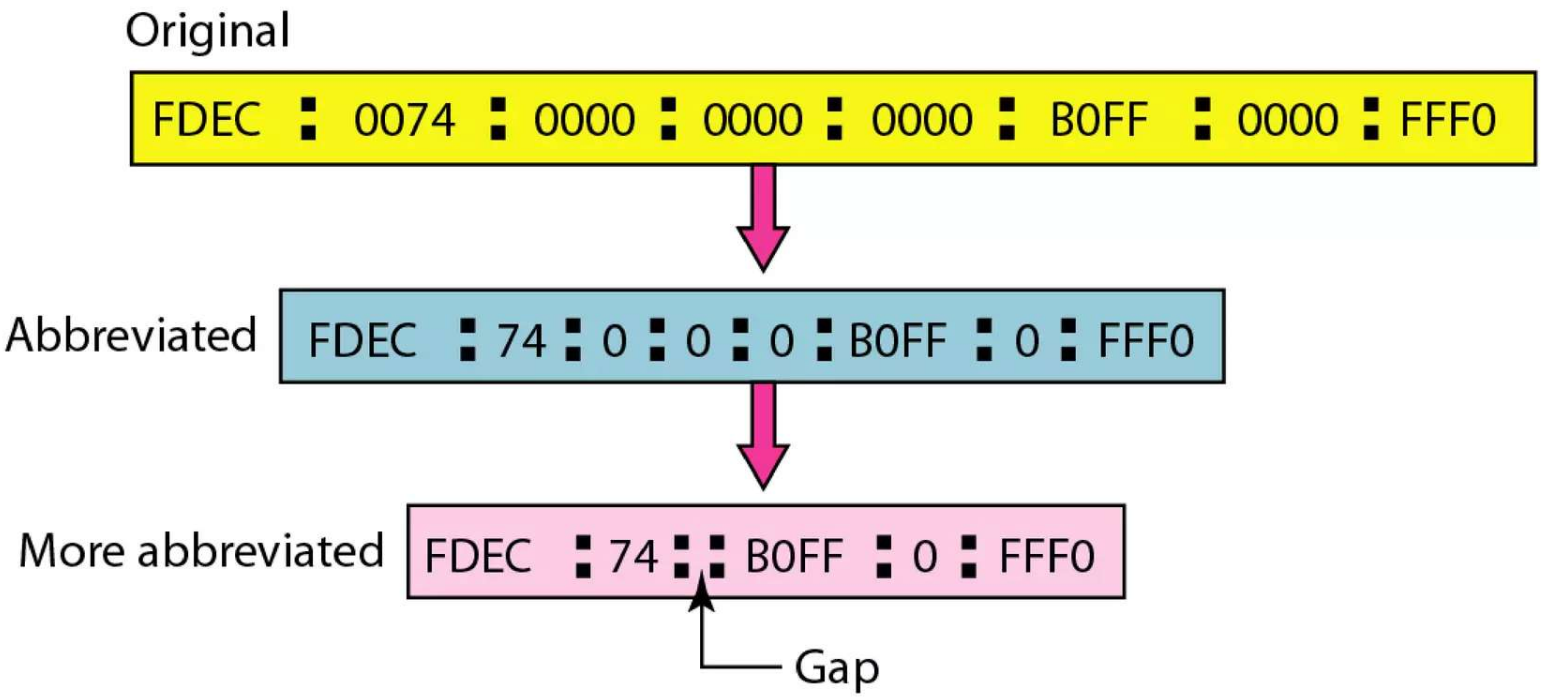


Figure 19.15 *Abbreviated IPv6 addresses*



Example 19.11

Expand the address 0:15::1:12:1213 to its original.

Solution

We first need to align the left side of the double colon to the left of the original pattern and the right side of the double colon to the right of the original pattern to find how many 0s we need to replace the double colon.

```
xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx
 0:   15:                   :    1:   12:1213
```

This means that the original address is.

```
0000:0015:0000:0000:0000:0001:0012:1213
```

Table 19.5 *Type prefixes for IPv6 addresses*

Type Prefix	*Type*	*Fraction*
0000 0000	Reserved	1/256
0000 0001	Unassigned	1/256
0000 001	ISO network addresses	1/128
0000 010	IPX (Novell) network addresses	1/128
0000 011	Unassigned	1/128
0000 1	Unassigned	1/32
0001	Reserved	1/16
001	Reserved	1/8
010	**Provider-based unicast addresses**	**1/8**

Table 19.5 *Type prefixes for IPv6 addresses (continued)*

Type Prefix	*Type*	*Fraction*
011	Unassigned	1/8
100	Geographic-based unicast addresses	1/8
101	Unassigned	1/8
110	Unassigned	1/8
1110	Unassigned	1/16
1111 0	Unassigned	1/32
1111 10	Unassigned	1/64
1111 110	Unassigned	1/128
1111 1110 0	Unassigned	1/512
1111 1110 10	Link local addresses	1/1024
1111 1110 11	Site local addresses	1/1024
1111 1111	Multicast addresses	1/256

Figure 19.16 *Prefixes for provider-based unicast address*

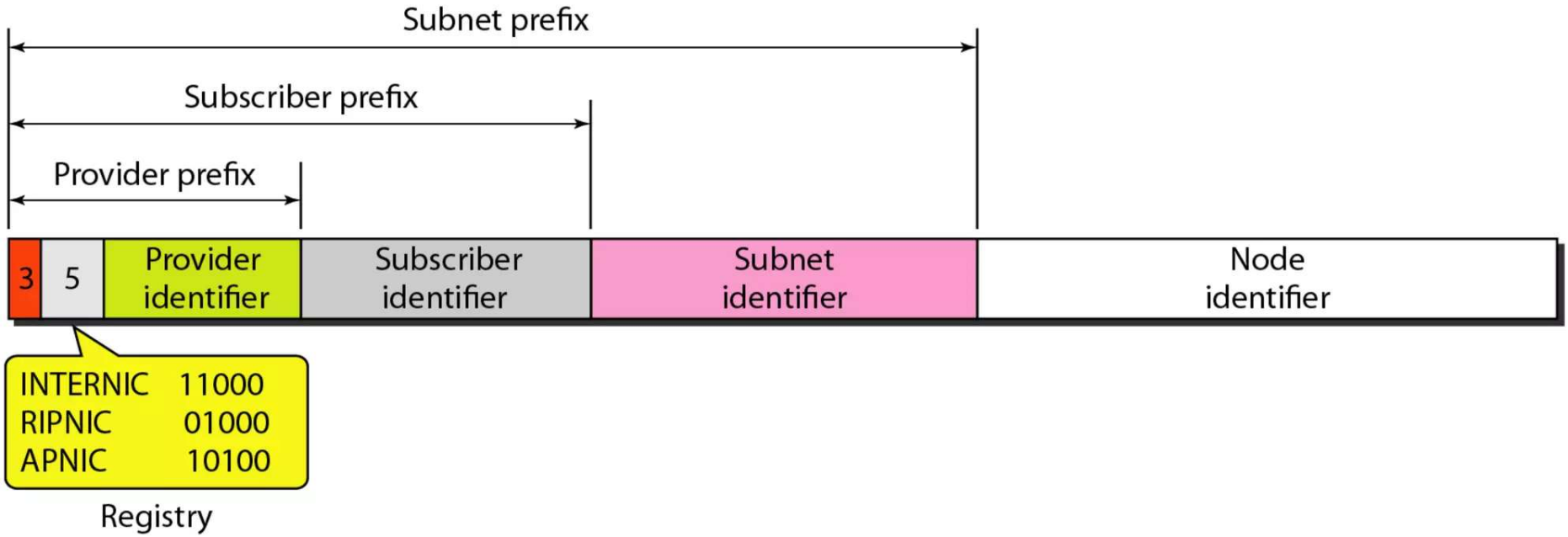


Figure 19.17 *Multicast address in IPv6*

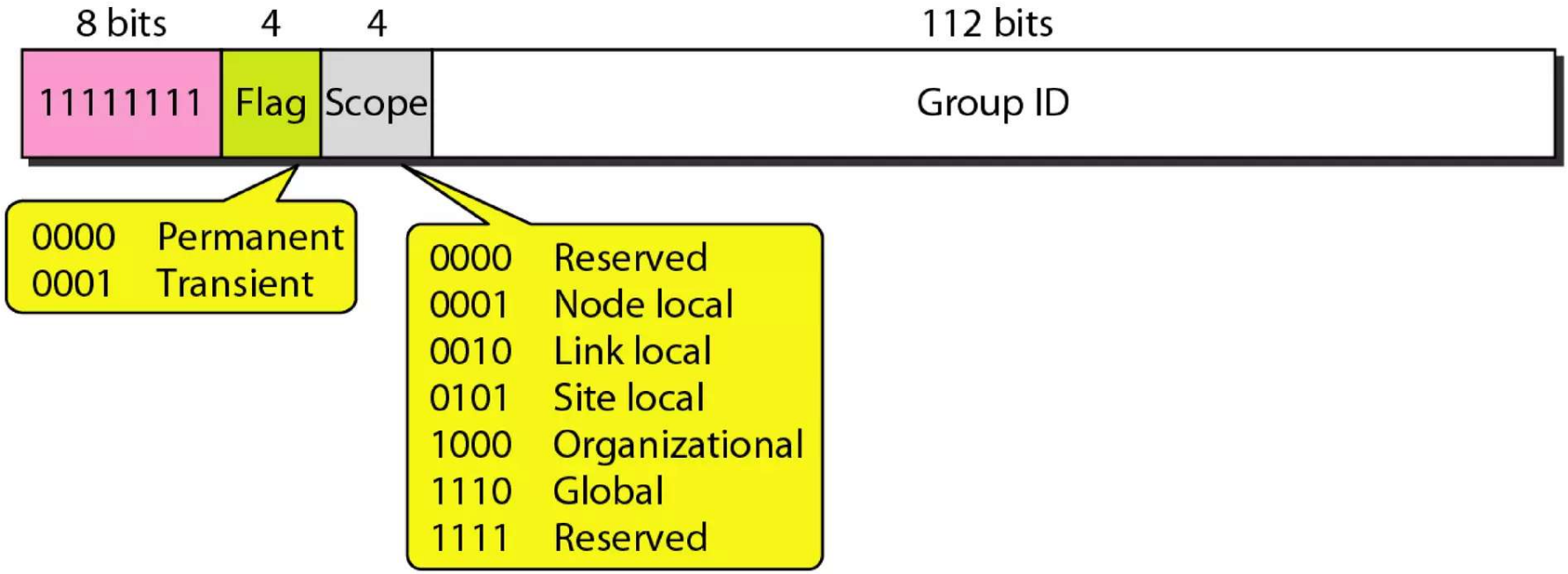


Figure 19.18 *Reserved addresses in IPv6*

